# A Multi-Agent Evidence Fusion Framework for Cross-Domain Misinformation Detection Using Affective Retrieval and External Evidence Verification

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# 1. Introduction

The rapid propagation of misinformation across online platforms has become one of the most significant challenges of the digital age. Social media ecosystems, online news portals, and citizen journalism platforms have collectively lowered the barrier to information publication, enabling the rapid dissemination of misleading, fabricated, or manipulated content across a global audience. The consequences of unchecked misinformation span domains including public health, electoral integrity, financial markets, and national security, making automated detection systems an urgent research priority.

Traditional misinformation detection systems have primarily relied on text-based features and semantic similarity measures to distinguish credible content from misleading claims. While these approaches have shown strong performance when training and test data originate from the same domain, the performance degrades across domains. For an instance a model trained on political misinformation, will frequently fail when applied to healthcare or financial misinformation because vocabulary, writing style, and contextual signals differ substantially. The impracticality of training a separate specialist model for every domain has driven research interest in **cross-domain misinformation detection**.

Recent advances in Large Language Models (LLMs) and retrieval-augmented generation (RAG) have opened new directions for cross-domain misinformation detection. Affect-aware retrieval methods have demonstrated that emotional patterns including fear, anger, uncertainty, and trust transfer more reliably across domains than surface-level semantic features. The RAEmoLLM framework (Liu et al., 2025) exemplifies this direction, demonstrating significant improvements over random and semantic retrieval baselines by leveraging affective embeddings extracted from an emotion-specialised language model.

However, current affect-aware retrieval systems remain limited by their dependence on a fixed retrieval database. When an emerging news event lacks closely related examples in the database, retrieval quality degrades, reducing misinformation detection accuracy. Furthermore, existing approaches typically rely on a single decision-making model, which reduces transparency and lacks in explaining how different evidence sources contribute to the final prediction. There is a need for a modular, explainable framework capable of integrating multiple heterogeneous evidence sources while explicitly handling uncertainty.

This paper presents a Multi-Agent Evidence Fusion Framework that addresses these limitations by extending the RAEmoLLM retrieval mechanism with two complementary analysis modules.

I. The first module implements affect-aware retrieval using affective embeddings and a logistic regression classifier trained on five emotional dimensions extracted by EmoLLaMA (Liu et al., 2024c).

II. The second module performs real-time external evidence retrieval using the Tavily Search API, obtaining supporting or contradicting evidence from live web sources. The outputs of these agents are subsequently combined to produce a final misinformation classification. The framework is evaluated on the FakeNewsNet dataset, comprising news articles from BuzzFeed and PolitiFact across two distinct domains.

The integration of machine learning and artificial intelligence into information verification systems represents a significant evolution from early rule-based approaches. Early automated fact-checking systems relied on manually engineered features such as bag-of-words representations, syntactic patterns, and handcrafted credibility heuristics. While these approaches established foundational vocabulary for the field, they were brittle and dependent on fixed feature sets that failed to generalise across writing styles and domains. The advent of deep learning, and subsequently transformer-based language models, fundamentally shifted the paradigm by enabling systems to learn rich contextual representations directly from raw text, capturing long-range dependencies and subtle semantic nuances that manual feature engineering could not.

Using Large Language Models for misinformation detection as proven well better where the relevant evidence is heterogeneous spanning emotional tone, factual consistency, source credibility, and social context and where no single signal is sufficient for reliable classification. The combination of Transformer Based mechanisms with Traditional Machine Learning represents the current frontier of the field, offering both improved accuracy and greater potential for interpretable, evidence-grounded explanations.

The move towards multi-agent and ensemble architectures further advances the determinism and reliability of AI-powered misinformation detection. Rather than delegating all analysis to a single monolithic model whose internal reasoning is opaque, multi-agent frameworks decompose the detection problem into discrete, auditable subtasks. Each agent operates within a well-defined scope affective analysis, external verification, or source consistency and produces a measurable output that can be independently inspected. This modular design reduces the risk of single-point failure: if one agent lacks sufficient evidence or produces a low-confidence output, the other agents can compensate through the fusion mechanism. The determinism introduced by explicit confidence scores, calibrated classifiers, and structured evidence representations contrasts sharply with the black-box nature of end-to-end neural systems, making the decision process traceable and the outputs more trustworthy in high-stakes deployment contexts.

# 2. Literature Survey

## 2.1 Background

The rapid growth of social media platforms and online news sources has led to an unprecedented increase in the spread of misinformation across multiple domains, including healthcare, politics, finance, and public policy. Traditional misinformation detection systems primarily rely on text-based features and semantic similarity to distinguish between real and fake information. However, these approaches often struggle when applied across domains because vocabulary, topics, and contextual information vary significantly between different areas.

To elaborate this challenge, consider a detection model formed for financial news verification. Using such a model on healthcare misinformation would be impractical due to domain-specific terminology and writing conventions. Conversely, training separate specialist models for every domain is computationally expensive and architecturally unsustainable. As a result, the research has been towards exploring retrieval-based approaches and Large Language Models that can leverage contextual examples to improve misinformation detection performance without domain-specific fine-tuning.

Recent studies have demonstrated that misinformation is not solely characterised by its textual content. Emotional characteristics, audience reactions, stance patterns, and supporting evidence often provide additional signals that help distinguish credible information from misleading claims. Affect-aware retrieval methods have gained attention because emotional patterns such as fear, anger, uncertainty, and trust frequently appear across different misinformation domains. Simultaneously, research in external evidence verification has demonstrated that real-time retrieval of supporting or contradicting web sources can substantially improve classification accuracy, particularly for emerging news events not represented in training data.

## 2.2 Review of Existing Literature

The following table summarises the key papers reviewed, their main contributions, methodological approaches, and relevance to the proposed framework.

| Paper | Main Idea | Key Prompts / Methods | Interesting Observation | Relevance |
|---|---|---|---|---|
| RAEmoLLM: Retrieval-Augmented Emotion-Aware Misinformation Detection (Liu et al., 2025) | Uses affective embeddings alongside semantic similarity for cross-domain retrieval without fine-tuning. | Template 1: Task + target text + few-shot examples + label. Template 2 adds explicit Vreg score appended to target text and each example. | Emotion patterns transfer better across domains than textual content. Vreg (sentiment strength) is the most discriminative single signal. | Forms the primary retrieval baseline for the proposed work. |
| Emotion-Based Misinformation Detection Survey (Liu et al., 2024d) | Reviews the role of emotions, comments, stance detection, and propagation behaviour in misinformation detection. | No specific prompts — surveys techniques across multiple models and datasets. | Audience reactions often contain useful credibility signals. Support and denial behaviour helps understand the confidence of a statement. | Motivates the use of social context and external evidence in the proposed framework. |
| EmoLLMs: Affective Large Language Models (Liu et al., 2024c) | Develops emotion-specialised language models for affective reasoning across five dimensions: EIreg, EIoc, Vreg, Voc, and Ec. | EIreg: assign intensity 0-1 per emotion. Vreg: assign valence 0-1. Voc: classify sentiment -3 to 3. Ec: detect which emotions are present. | Emotion-aware models outperform generic LLMs on affective tasks. The last hidden layer vector under emotional prompting encodes richer affective signal. | Provides the foundation for extracting both explicit affective scores and implicit affective embeddings used in retrieval. |
| FANG: Social Context for Fake News Detection | Utilises social interactions and surrounding context for fake news detection via | No specific prompts — graph-based approach modelling user- | Social behaviour provides useful credibility indicators. Network structure | Supports the incorporation of social context and source credibility signals. |

| Paper | Main Idea | Key Prompts / Methods | Interesting Observation | Relevance |
|---|---|---|---|---|
| | graph neural networks. | news-source relationships. | reveals patterns invisible in text alone. | |
| FactLLaMA: Optimizing Instruction-Following LLMs with External Knowledge (Cheung & Lam, 2023) | Combines LLaMA with Google Search API evidence retrieval for fact-checking without relying on internal model knowledge alone. | Instruction-Evidence-Input format: Instruction + Evidence (retrieved via Google API) + Input claim. Fine-tuned with LORA. | External knowledge consistently improves performance over instruction-tuning alone (F1: 0.5565 vs 0.5376 on RAWFC). Model struggles most with partially true labels. | Directly motivates the external evidence verification agent. Shows that grounding LLM decisions in retrieved evidence produces measurable gains. |
| Re-Search for the Truth — STEEL (Li et al., 2024) | Multi-round internet-based evidence retrieval with LLM reasoning and confidence-based re-search triggers. | Step-check prompt: respond 0 (true), 1 (false), or 2 (NEI). If NEI: output missing info + updated queries + compressed evidence. Re-search triggered when confidence < 50%. | Multi-round retrieval outperforms single-step search across all datasets. Performance peaks at 3 retrieval rounds. Source credibility filtering using blacklist of 1,044 known fake news websites. | Informs the web verification agent design. NEI handling and confidence scoring are directly relevant to the proposed framework. |

## 2.3 Synthesis

Reading across the reviewed literature, several complementary themes emerge. RAEmoLLM establishes that affective embeddings extracted under emotion-specific prompt contexts provide a stronger retrieval signal than generic semantic embeddings, with Vreg (sentiment valence strength) identified as the most discriminative single dimension for fake news detection. The EmoLLMs paper provides the model and prompting methodology used to extract these affective dimensions, including five categories: EIreg (emotion intensity regression), EIoc (emotion intensity classification), Vreg (valence strength), Voc (valence classification), and Ec (emotion detection). Together, these two works form the affective foundation of the proposed framework.

FactLLaMA and STEEL both demonstrate that grounding LLM reasoning in externally retrieved evidence consistently improves detection accuracy, particularly for claims not well represented in static training databases. STEEL further shows that a multi-round retrieval strategy with explicit confidence scoring and re-search triggers outperforms single-step retrieval. This motivates the external evidence verification agent in the proposed framework, which follows a two-chain architecture separating query generation from verdict reasoning.

The social context literature represented by FANG highlights that source credibility and network structure provide signals that complement text-based and affect-based analysis. While the proposed framework does not implement full graph-based social modelling, the source credibility signal derived from publisher-level labels in the FakeNewsNet dataset serves as a lightweight substitute.

### 2.4 Proposed Solution:

Following a careful analysis of the existing literature, several targeted modifications to the established RAEmoLLM pipeline can make the overall misinformation detection process more deterministic, measurable, and interpretable. Rather than relying entirely on the in-context reasoning capability of a large language model which introduces variability across inference calls and lacks a calibrated confidence output — the proposed framework introduces two complementary analytical components that produce quantifiable soft labels. These components operate as independent evidence sources whose outputs are subsequently combined through an ensemble fusion mechanism.

The first modification replaces the direct in-context learning classification with a supervised logistic regression model trained on affective features. Using EmoLLaMA-chat-7B, five affective scores are extracted per article: valence strength (Vreg), sentiment classification (Voc), and emotion intensities for anger, fear, joy, and sadness (EIreg). These scores, combined with the ground-truth labels of the training dataset, provide a structured five-dimensional feature space on which a logistic regression model can be trained. Logistic regression was selected deliberately over more complex alternatives because its linear decision boundary is well-suited to the modest dataset size, its coefficients are directly interpretable as feature importance weights, and its output probability is a calibrated confidence score rather than a hard binary prediction. The outcome of this first pipeline phase is therefore not a definitive fake or real label but a soft label — a continuous probability value indicating how strongly the affective profile of the sample article aligns with the affective patterns previously observed in fake versus real news. This soft label is measurable, reproducible across inference runs, and directly grounded in the statistical properties of the training data.

The second modification introduces a function-calling LLM agent responsible for gathering real-time external evidence from the internet. Modern large language models, including those in the Mistral, Llama, and Qwen families, support structured function calling, enabling them to invoke external tools within a reasoning pipeline. Search APIs such as Tavily and SerpAPI expose internet search capabilities that can be called as tools by the LLM, returning structured results — including source URLs, page titles, and text snippets — that the model can process to derive meaningful analytical insight. The proposed agent uses this capability to retrieve relevant articles, fact-checks, and corroborating or contradicting sources for the sample article under observation. Rather than producing a hard classification, the agent analyses the retrieved evidence against the original article and assigns a confidence score reflecting the degree to which the external sources support or contradict the claim. This produces a second soft label — an evidence consistency score — that captures the factual alignment between the article and independently retrieved web sources.

Taken together, the two components form an ensemble pattern in which independent analytical channels each contribute a soft confidence score. The affective channel captures emotional and sentiment-level signals that characterise the writing style and tonal patterns of the article, while the evidence channel captures factual consistency with the broader information landscape at the time of evaluation. The final prediction is obtained by combining these scores through a fusion mechanism, producing an overall confidence value and a final fake or real classification. This ensemble design ensures that no single evidence source dominates the prediction, that each component can be independently inspected and audited, and that the system degrades gracefully when one channel produces low-confidence output — for example, when external evidence is insufficient or when the affective profile of an article falls near the classification boundary.

## 3. Identified Research Gap

The literature indicates that affect-aware retrieval significantly improves cross-domain misinformation detection by selecting emotionally relevant examples. However, most existing retrieval frameworks primarily focus on semantic and affective similarity while overlooking other forms of contextual information that may also transfer across domains. Although several studies have demonstrated the importance of audience reactions, stance patterns, and credibility signals, these factors are typically investigated separately from retrieval-based approaches.

Another critical limitation of current retrieval-based misinformation detection systems is their dependence on information available within the training dataset. When a news event is recent or concerns a niche topic, closely related examples may not exist in the retrieval database, reducing the effectiveness of affective retrieval methods. This static dependence on training data stands in contrast to the dynamic nature of misinformation, which continuously evolves across new events and narratives.

Furthermore, most existing approaches rely on a single decision-making model, which limits transparency regarding how different evidence sources contribute to the final prediction. When a system produces a fake or real label without decomposing the evidence, practitioners cannot assess the relative contribution of emotional similarity, factual consistency, or source credibility. This opacity reduces trust in automated predictions and limits the interpretability of the system.

Therefore, there remains an opportunity to investigate a modular framework that combines affect-aware retrieval with dynamic external evidence retrieval and mathematically grounded evidence fusion. Such a framework would allow independent analysis of multiple information sources while providing a more transparent and explainable decision-making process that does not require complex knowledge graph construction or domain-specific fine-tuning.

# 4. Proposed Methodology

## 4.1 Overview

Based on the identified research gap, this work proposes a Multi-Agent Evidence Fusion Framework for cross-domain misinformation detection. The proposed system extends the RAEmoLLM framework by incorporating multiple specialised agents that independently analyse different aspects of a news article. Rather than relying solely on a Large Language Model to generate the final prediction, each agent produces measurable evidence that is subsequently fused to provide a mathematically grounded and explainable decision process.

The framework comprises three principal agents operating in parallel on a given input article, followed by a structured evidence fusion step that combines their outputs into a final classification. The three agents address complementary dimensions of misinformation analysis: affective characteristics, external evidence consistency, and claim-source structure.

## 4.2 Agent 1 — Affective Retrieval Agent (RAEmoLLM)

The first agent implements the RAEmoLLM framework extended with a logistic regression classifier. Given an input news article, the agent first extracts the full affective representation by computing five emotional dimensions using EmoLLaMA-chat-7B: Vreg (valence strength, 0–1), Voc (valence classification, –3 to 3), EIreg for anger, fear, joy, and sadness (each 0–1), and Ec (emotion detection). These scores are extracted using the exact prompt templates prescribed by the EmoLLMs paper, ensuring that the hidden layer representations are elicited under affective reasoning context.

In parallel, an implicit affective embedding (4096 dimensions) is extracted from the last hidden layer of EmoLLaMA while processing the Vreg prompt. This embedding captures the affective state of the model when reasoning about the emotional valence of the text, providing a richer representation than a plain text embedding. These embeddings are stored in a retrieval database indexed alongside their ground-truth labels.

For a given target article, the agent retrieves the top-K most emotionally similar articles from the source domain database using cosine similarity in the 4096-dimensional affective embedding space. This leave-one-domain-out retrieval strategy — retrieving from source domains and evaluating on the target domain — constitutes the cross-domain component of the framework. The retrieved examples and their labels are used as few-shot demonstrations in a Template 2 prompt (Vreg score appended to each example) fed to a local Mistral-7B model for in-context classification.

Additionally, a logistic regression classifier trained on the five explicit affective dimensions (vreg, eireg_anger, eireg_fear, eireg_joy, eireg_sadness) provides a complementary confidence score. This classifier, trained on the full labelled dataset using five-fold stratified cross-validation, produces a probability estimate reflecting how closely the affective profile of the target article resembles previously observed fake or real news. The logistic regression output is stored as the affective confidence score feeding into the evidence fusion step.

### 4.3 Agent 2 — External Evidence Verification Agent

The second agent performs real-time external evidence retrieval to verify or contradict the claims presented in the input article. Rather than relying solely on the static retrieval database, this agent queries the Tavily Search API to obtain up-to-date evidence from live web sources, addressing the static database limitation identified in the research gap.

The agent operates as a two-chain pipeline. In Chain 1, a large language model — Qwen2.5-72B-Instruct accessed via the HuggingFace Inference API — receives the article title and generates a focused, concise search query optimised for web retrieval. This step prevents the literal article title, which may contain sensational or misleading language, from being used directly as a search query, which could bias the retrieved results. The generated query is submitted to the Tavily Search API with advanced search depth, retrieving up to ten relevant results alongside source URLs, page titles, and text snippets.

In Chain 2, a second LLM call presents the retrieved evidence snippets alongside the original article title and requests a structured JSON verdict. The verdict includes a classification of SUPPORTED, CONTRADICTED, or INSUFFICIENT_EVIDENCE; a confidence score between 0 and 1; a one-sentence explanation citing the key evidence; and lists of supporting and contradicting source URLs. This two-chain separation of retrieval and reasoning ensures that the model performing verdict generation receives a clean evidence dossier rather than needing to simultaneously manage tool calling and classification reasoning.

The output of this agent — specifically the verdict category and confidence score — contributes to the evidence fusion step as a quantitative measure of factual consistency between the article and independently retrieved web sources.

### 4.4 Evidence Fusion

After both agents complete their independent analyses, their outputs are combined to produce the final misinformation classification. The affective retrieval agent produces two signals: the in-context LLM classification from the RAEmoLLM pipeline and the logistic regression confidence score from the affective classifier. The external evidence agent produces the web verdict and its associated confidence score.

These signals are combined using a weighted evidence fusion approach. The affective confidence score indicates the degree to which the emotional profile of the article resembles known misinformation. The web verdict confidence indicates the degree to which independent external sources support or contradict the article. When these signals are in agreement — for example, a high affective fake confidence combined with a CONTRADICTED web verdict — the combined evidence provides strong grounds for a fake classification. When signals conflict, the framework appropriately flags lower confidence, reflecting the inherent ambiguity.

The final classification is presented alongside an interpretable evidence summary that decomposes the contribution of each agent, enabling practitioners to understand why a particular article received its label. This transparency addresses the explainability gap identified in the research gap section.

## 4.5 Evaluation Strategy

The proposed framework will be evaluated by comparing its performance against three baselines: random retrieval, semantic retrieval using RoBERTa embeddings, and affect-aware retrieval using the original RAEmoLLM framework. Performance will be measured using weighted Accuracy, Precision, Recall, and F1 score, following the evaluation protocol established by Liu et al. (2025). The leave-one-domain-out strategy will be applied across the BuzzFeed and PolitiFact domains in the FakeNewsNet dataset, treating each domain alternately as the target domain.

# 5. Implementation

## 5.1 Dataset and Preprocessing

The FakeNewsNet dataset (Shu et al., 2020) serves as the primary evaluation benchmark. The dataset comprises news articles collected from two distinct fact-checking platforms: BuzzFeed and PolitiFact. Both platforms focus on political news, providing two domains for cross-domain evaluation under the leave-one-domain-out strategy. The dataset was obtained via Kaggle in CSV format, comprising four files: BuzzFeed fake, BuzzFeed real, PolitiFact fake, and PolitiFact real news content.

Preprocessing involved removing columns not relevant to affective or textual analysis, including article identifiers, thumbnail images, author metadata, publication dates, canonical links, and embedded multimedia references. The remaining columns — title, article body text, URL, source publisher, and metadata descriptions — were retained. A unified full_text field was constructed by concatenating the article title and body text for each record, as the combined representation provides richer affective signal than either component alone. Rows where the resulting full_text field was empty or contained fewer than twenty characters were discarded as uninformative.

Following preprocessing, the dataset comprises 422 articles with a balanced distribution of 211 fake and 211 real articles split equally across the two source publishers. Publisher identity was encoded as a domain label (BuzzFeed or PolitiFact) to support leave-one-domain-out cross-domain evaluation. A binary label column was created mapping real to 0 and fake to 1 for classifier training.

## 5.2 Affective Scoring with EmoLLaMA

Affective scoring was performed using EmoLLaMA-chat-7B, quantised to Q4_K_M precision in GGUF format for local inference on Apple Silicon hardware via the llama-cpp-python library with Metal GPU acceleration. The model was loaded with a 2048-token context window with all available GPU layers utilised through the n_gpu_layers=-1 parameter, consuming approximately 7-8 GB of unified memory on the target M4 MacBook Pro with 16GB RAM.

Five affective dimensions were extracted per article using the exact prompt templates prescribed by the EmoLLMs paper. For Vreg (valence strength), the model was prompted to assign a real-valued score between 0 (most negative) and 1 (most positive) representing the overall sentiment intensity of the text. For EIreg, four separate inference calls were made per article requesting intensity scores for anger, fear, joy, and sadness individually. For Voc (valence classification), the model was prompted to categorise the text into one of seven ordinal classes ranging from -3 (very negative) to +3 (very positive). For Ec (emotion detection), the model was prompted to identify which of eleven emotional categories were present in the text.

A robust generation function was implemented with progressive text truncation as a fallback mechanism. If an inference call raised a RuntimeError due to context overflow — indicated by the llama_decode returned -3 error code — the text was progressively shortened from 250 to 150 to 80 characters before reattempting the call. Each affective dimension was extracted and parsed individually, with float values clamped to the [0, 1] range and integer values clamped to the [-3, 3] range for Voc. The extraction process was checkpointed every 25 articles to enable resumption after interruption.

The complete affective scoring pipeline processed all 422 articles sequentially, producing the fakenewsnet_with_affect.csv output file containing the original article metadata alongside the seven extracted affective scores per article.

## 5.3 Affective Embedding Extraction

Implicit affective embeddings were extracted from the last hidden layer of EmoLLaMA to support cosine similarity retrieval in the RAEmoLLM pipeline. The model was reloaded in embedding mode by setting the embedding=True parameter in the Llama constructor, switching the inference backend from generation to embedding extraction.

Each article was embedded by passing it through the Vreg prompt template before extraction. This approach, prescribed by the EmoLLMs paper, ensures that the 4096-dimensional hidden layer vector captures the model internal affective reasoning state about sentiment valence rather than a generic text representation. The resulting vectors provide richer cross-domain retrieval signal than plain semantic embeddings because they encode emotional context rather than topical similarity.

A memory-efficient disk-writing strategy was employed to handle the 422 × 4096 float32 embedding matrix without loading the full array into RAM. The output file was pre-allocated on disk using NumPy memory-mapped arrays (numpy.lib.format.open_memmap), and each embedding was written to disk immediately after extraction before releasing the memory reference. This approach maintains a constant RAM footprint of approximately 16 kilobytes per iteration regardless of dataset size. A plain-text progress file tracked the last completed row index, enabling automatic resumption following interruptions.

The final affective_embeddings.npy file contains a (422, 4096) float32 array. Token-level outputs from the embedding API, which return shape (N_tokens, 4096), were mean-pooled across the token dimension before storage to obtain a single sentence-level vector per article.

## 5.4 Affective Analysis

Prior to classifier training, a comprehensive affective analysis was conducted to characterise the distributional differences between fake and real news in the affective feature space, replicating the analysis methodology of Liu et al. (2025).

Table 1 presents the mean affective scores and t-test statistics for each dimension across the fake and real classes. Two dimensions reached statistical significance at $p < 0.05$: EIreg-Anger ($t = -4.58$, $p = 6.04\times10^{-6}$) and EIreg-Fear ($t = -2.51$, $p = 0.013$). Fake news articles exhibited significantly higher anger intensity (mean: 0.436) than real articles (mean: 0.408), and significantly higher fear intensity (0.453 vs 0.434). These findings are directionally consistent with the RAEmoLLM paper findings on the AMTCele dataset, where fake news also showed significantly elevated anger and fear scores. Sentiment strength (Vreg), joy, and sadness did not reach significance, suggesting that affective signals alone are insufficient for reliable classification and motivating the multi-agent fusion approach.

Examining high-intensity patterns, 19.0% of fake articles exceeded the 0.5 threshold for anger compared to 7.6% of real articles — a 2.5-fold difference. For fear, 26.5% of fake articles showed high intensity versus 17.1% of real articles. Anger was the dominant emotion in 54.5% of fake articles and 47.9% of real articles, while fear was the dominant emotion in a higher proportion of real articles (43.6%) than fake (38.4%). These distributional patterns provide interpretable evidence that affective signals carry discriminative information for misinformation detection, even if they alone are insufficient for high-accuracy classification.

## 5.5 Logistic Regression Classifier

A logistic regression classifier was trained on the five explicit affective dimensions to produce a calibrated probability estimate for the affective confidence score. Logistic regression was selected as the most appropriate model for this component given the small dataset size of 422 articles and the five-dimensional feature space. More complex models including gradient boosting and support vector machines were considered but risk overfitting with limited data. The linear decision boundary of logistic regression is well-suited to the modest and statistically supported separability observed in the affective analysis.

The classifier was implemented as a scikit-learn Pipeline comprising a StandardScaler for feature normalisation followed by a LogisticRegression estimator with L2 regularisation (C=1.0, max_iter=1000). Training and evaluation used five-fold stratified cross-validation to preserve class balance across folds, providing an unbiased estimate of generalisation performance on the 422-sample dataset. The full pipeline was subsequently fitted on the complete dataset and serialised to disk using joblib for use during inference.

The trained classifier provides a predict_from_affect function that accepts a dictionary of affective scores and returns a predicted label (fake or real) alongside a calibrated confidence score reflecting the probability assigned by the logistic regression model. This confidence score constitutes the affective agent output that feeds into the evidence fusion step.

The feature weights learned by the classifier are interpretable and consistent with the affective analysis findings. EIreg-Anger and EIreg-Fear carry the largest absolute coefficients in the positive direction (towards fake), while Vreg and EIreg-Joy carry negative coefficients (towards real), aligning with the statistically significant differences observed in the t-test analysis.

## 5.6 External Evidence Verification

The external evidence verification agent was implemented as a two-chain LangChain pipeline using Qwen2.5-72B-Instruct accessed via the HuggingFace Inference API with automatic provider selection. The choice of a 72-billion parameter instruction-tuned model reflects the complexity of the reasoning task: evaluating multiple web evidence snippets against a news claim requires strong language understanding and structured output generation.

Chain 1 handles query generation and evidence retrieval. The LLM receives a system prompt instructing it to generate a single concise web search query optimised to find verifying or debunking evidence for the given article. This indirect query generation step prevents literal use of the article title as a search query, which could embed sensational or misleading language into the search and bias the retrieved results. The generated query is submitted to the Tavily Search API with advanced search depth, retrieving up to ten results with full content snippets.

Chain 2 handles verdict generation. The LLM receives a system prompt specifying the exact JSON output format required, followed by the article title, Tavily own answer summary, and the full evidence text from all retrieved sources. The model is instructed to output only valid JSON containing a verdict field (SUPPORTED, CONTRADICTED), a confidence score, a one-sentence explanation, and lists of supporting and contradicting source URLs. Strict JSON-only output was specified in the system prompt to prevent LLM tendency to prepend conversational preamble that breaks downstream parsing.

A robust JSON extraction function processes the raw LLM output by locating the first and last curly brace positions and attempting to parse the enclosed substring. If parsing fails, the verdict defaults to INSUFFICIENT_EVIDENCE with a neutral confidence of 0.5, ensuring the pipeline degrades gracefully. All verification results are checkpointed every ten articles and written to web_verification_results.csv for subsequent use in the evaluation pipeline.

## 5.7 RAEmoLLM Retrieval and Inference Pipeline

The complete RAEmoLLM pipeline implements the leave-one-domain-out evaluation strategy prescribed by Liu et al. (2025). For each target article, the pipeline retrieves the top-4 most emotionally similar articles from the source domain — the domain not currently serving as the target — using cosine similarity computed in the 4096-dimensional affective embedding space. The sklearn.metrics.pairwise.cosine_similarity function operates on the memory-mapped embedding file without loading the full matrix into RAM.

Prompts are constructed following Template 2 from the RAEmoLLM paper, which appends the explicit Vreg score to each retrieved example and to the target article. The proposed framework extends this template by additionally including the web verification verdict and confidence score as a separate evidence line. This allows the inference LLM to weight the web evidence alongside the affective retrieval examples when generating its classification decision. The task prompt instructs the model to classify the target text as 0 (Fake) or 1 (Legit), and the raw output is parsed to extract the predicted label.

Inference is performed using a locally deployed Mistral-7B-Instruct model in Q4_K_M GGUF format, providing fast inference without external API costs. The model output is parsed with a simple rule-based parser: responses containing "1", "legit", or "real" are mapped to the real class, and all other responses default to the fake class. This conservative defaulting strategy reflects the relative cost asymmetry between false negatives and false positives in misinformation detection.

## 5.8 Implementation Environment

All components were implemented in Python 3.11 within a conda virtual environment. EmoLLaMA inference used llama-cpp-python compiled with Metal GPU acceleration for Apple Silicon (CMAKE_ARGS="-DGGML_METAL=on"). Web verification used LangChain-HuggingFace for model loading, LangChain-Community for the Tavily search tool, and the Tavily Python client for direct API access. The scikit-learn library provided the logistic regression classifier, standardisation pipeline, and evaluation metrics. All file I/O used pandas for tabular data and NumPy for embedding arrays.

# 6. Evaluation

## 6.1 Experimental Setup

The evaluation follows a cross-dataset generalisation protocol, a design choice deliberately distinct from the standard train-test split approach. The logistic regression affective classifier and affective embeddings were trained exclusively on the FakeNewsNet dataset comprising 422 articles sourced from BuzzFeed and PolitiFact. The evaluation was conducted on the FakeNewsAMT dataset, a completely separate benchmark used by Liu et al. (2025) as one of their three primary evaluation sets. FakeNewsAMT comprises 480 articles spanning six distinct domains — business, education, entertainment, politics, sports, and technology — with 40 fake and 40 real articles per domain, providing a balanced and domain-diverse test set. No articles from FakeNewsAMT were seen during training of any component of the proposed framework, making this a zero-shot cross-dataset generalisation evaluation. This design is more rigorous than a held-out split from the same dataset, as it directly assesses whether the learned affective patterns and web verification signals transfer to an entirely different data source and domain distribution.

The web verification agent was evaluated using two LLM configurations to assess the sensitivity of the pipeline to the choice of reasoning model. The first configuration used Meta-Llama-3.3-70B-Instruct accessed via the Groq API, selected for its generous free-tier rate limits and strong instruction-following capability. The second configuration used Qwen2.5-72B-Instruct accessed via the HuggingFace Inference API with automatic provider selection. Both models received identical system prompts and chain architectures, allowing direct comparison of their verdict generation quality under the same evidence inputs.

Performance is reported using weighted Accuracy, Precision, Recall, and F1 score across the full FakeNewsAMT test set and broken down per domain. Comparison is made against the RAEmoLLM paper results on the AMTCele dataset, which combines FakeNewsAMT with a Celebrity domain extension. Since the proposed evaluation uses only the FakeNewsAMT component, comparisons to RAEmoLLM domain-level results are directional rather than exact.

## 6.2 Overall Results

~~Table 2~~ presents the overall performance of the proposed framework against the RAEmoLLM baseline reported by Liu et al. (2025). Results for the Qwen2.5-72B configuration will be populated upon completion of the full experimental run.

| **Metric** | **Proposed Framework (Llama-3.3-70B)** | **Proposed Framework (Qwen2.5-72B)** | **RAEmoLLM Mistral-7b Template 1 (Vreg)** | **RAEmoLLM Mistral-7b Template 2 (Vreg + explicit)** |
|---|---|---|---|---|
| Accuracy | **0.8655** | — | 0.7404 | 0.7717 |
| F1 (weighted) | **0.8651** | — | 0.7404 | 0.7717 |
| Precision (weighted) | **0.8699** | — | — | — |
| Recall (weighted) | **0.8655** | — | — | — |

The proposed framework using Llama-3.3-70B achieves a weighted F1 of 0.8651, outperforming the best reported RAEmoLLM result on the AMTCele benchmark (F1 = 0.7717, Template 2 with explicit Vreg) by 8.9 percentage points. This improvement is particularly notable given that the proposed framework does not use in-context retrieval from the test domain — all classification signals derive from a classifier trained on a completely separate dataset and from live web evidence retrieved at inference time. Four articles produced pipeline errors during evaluation due to API timeouts and were excluded from metric computation, representing an error rate of 0.83% across the 480-article test set.

## 6.3 Per-Domain Results

~~Table 3~~ presents the domain-level breakdown of accuracy and F1 score for the proposed framework, enabling comparison against the leave-one-domain-out results reported by Liu et al. (2025).

| **Domain** | **N** | **Proposed Acc** | **Proposed F1** | **Correct/Total** | **RAEmoLLM approx. F1** |
|---|---|---|---|---|---|
| Business | 80 | 0.8250 | 0.8246 | 66/80 | ~0.72 |
| Education | 77 | 0.7662 | 0.7646 | 59/77 | ~0.71 |
| Entertainment | 79 | 0.8608 | 0.8595 | 68/79 | ~0.75 |
| Politics | 80 | 0.8875 | 0.8871 | 71/80 | ~0.78 |
| Sports | 80 | **0.9500** | **0.9499** | 76/80 | ~0.76 |
| Technology | 80 | 0.9000 | 0.8990 | 72/80 | ~0.74 |
| **Overall** | **476** | **0.8655** | **0.8651** | **412/476** | **~0.7717** |

The proposed framework outperforms the RAEmoLLM baseline across all six domains. Sports achieves the highest accuracy at 0.9500, likely attributable to the highly factual and verifiable nature of sports claims — match results, statistics, and player transfers are readily corroborated or contradicted by web search. Education is the weakest domain at 0.7662, consistent with the RAEmoLLM paper which also reports lower performance on education content, reflecting the more nuanced and opinion-adjacent nature of educational news claims that are harder for web evidence to clearly support or contradict.

### 6.4 Fake vs Real Detection Asymmetry

| Class | Precision | Recall | F1 | Detection Rate | Correct/Total |
|---|---|---|---|---|---|
| Real | 0.83 | 0.92 | 0.87 | 92.1% | 220/239 |
| Fake | 0.91 | 0.81 | 0.86 | 81.0% | 192/237 |

The framework detects real news more reliably (92.1%) than fake news (81.0%). This asymmetry is expected and interpretable: real articles are more likely to be corroborated by multiple independent web sources, producing SUPPORTED verdicts with high confidence that clearly signal legitimate content. Fake articles present a more challenging detection problem — some are constructed to closely mirror real reporting styles, contain partially true claims, or concern niche events for which web evidence is sparse. In these cases the web verification agent returns INSUFFICIENT_EVIDENCE, defaulting to a neutral confidence score of 0.5 that weakens the fake signal in the fusion step.

### 6.5 Agent Contribution Analysis

To assess the independent contribution of each pipeline component, the affective classifier and web verification signals were analysed separately.

*Table 5: Agent-level contribution analysis — Proposed Framework (Llama-3.3-70B)*

| Component | F1 (weighted) | Notes |
|---|---|---|
| Affect classifier only (logistic regression) | ~0.56 | 5-fold CV on FakeNewsNet training set |
| Web verification only (CONTRADICTED = fake) | ~0.71 | Articles with definitive verdicts only |
| Combined (weighted fusion 40% affect + 60% web) | **0.8651** | Full test set |
| RAEmoLLM best baseline | 0.7717 | Liu et al. (2025), AMTCele |

The affective classifier alone achieves approximately F1 = 0.56 in cross-validation on the training set, consistent with the t-test analysis showing that only anger and fear reach statistical significance as discriminative features. The web verification signal alone, evaluated only on articles receiving definitive SUPPORTED or CONTRADICTED verdicts, achieves approximately F1 = 0.71. The weighted fusion of both signals produces F1 = 0.8651, demonstrating that the ensemble combination substantially outperforms either component independently. This result validates the core research contribution: neither affective classification nor web verification alone is sufficient, but their combination provides performance exceeding the RAEmoLLM baseline by a significant margin.

## 6.6 Error Analysis

Examining the 64 misclassified fake articles and 19 misclassified real articles reveals consistent patterns. The majority of fake articles misclassified as real received SUPPORTED web verdicts — the web agent found corroborating sources because the fake article was based on a real event with fabricated details, causing the search to retrieve legitimate articles about the underlying event rather than debunking sources. This is particularly evident in the education domain, where fake articles frequently describe real schools or policies with exaggerated or false statistics that are difficult to contradict via short evidence snippets.

Misclassified real articles predominantly occurred when the article discussed a niche or region-specific topic for which the Tavily search returned limited evidence, producing INSUFFICIENT_EVIDENCE verdicts, and when the affective profile of the article happened to resemble the fake news distribution — particularly articles using strongly negative or fear-laden language about legitimate crises or controversies.

These error patterns directly motivate the future work described in Section 7, including threshold tuning to improve fake recall and the incorporation of source credibility blacklists to help the web verification agent distinguish between corroboration of the underlying event and corroboration of the specific claim.

## 6.7 Comparison Summary

The proposed framework demonstrates consistent and substantial improvements over the RAEmoLLM baseline across all measured dimensions. The 8.9 percentage point improvement in F1 score on a completely unseen test dataset, achieved without any domain-specific fine-tuning or in-context examples from the test domain, provides strong evidence that the combination of affective classification and real-time web verification constitutes a meaningful contribution to cross-domain misinformation detection. The results further confirm that the modular multi-agent design enables more reliable detection than single-signal approaches, with each agent contributing independently measurable value that compounds through the fusion mechanism.

# 7. Conclusion

This paper has presented a Multi-Agent Evidence Fusion Framework for cross-domain misinformation detection that extends the RAEmoLLM affective retrieval mechanism with two complementary components: a logistic regression affective classifier and a two-chain web evidence verification pipeline. The framework addresses three limitations identified in the existing literature: the static database dependence of affect-aware retrieval, the lack of real-time external evidence grounding, and the opacity of single-model decision-making.

The affective analysis conducted on the FakeNewsNet dataset replicates and extends the findings of Liu et al. (2025), confirming that anger ($p = 6.04 \times 10^{-6}$) and fear ($p = 0.013$) are significantly elevated in fake news articles compared to real news. These findings validate the use of affective signals as discriminative features and motivate the logistic regression classifier as a lightweight, interpretable component producing calibrated confidence scores from the five-dimensional affective feature space.

The web evidence verification agent addresses the static database limitation by dynamically retrieving and evaluating real-time evidence from live web sources. The two-chain architecture, separating query generation from verdict reasoning, improves the reliability of the evidence assessment compared to single-call approaches and is directly informed by the STEEL and FactLLaMA methodologies reviewed in the literature.

Future work will explore the application of Dempster-Shafer Evidence Theory as a principled fusion mechanism to replace the current weighted combination, enabling explicit uncertainty modelling when agent outputs conflict. Extension to additional datasets including PHEME (Twitter rumours) and COCO (COVID-19 conspiracy theories) will assess the generalisability of the framework across misinformation types. The incorporation of source credibility signals derived from known fake news website blacklists represents a natural extension of the current claim-source consistency analysis.

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